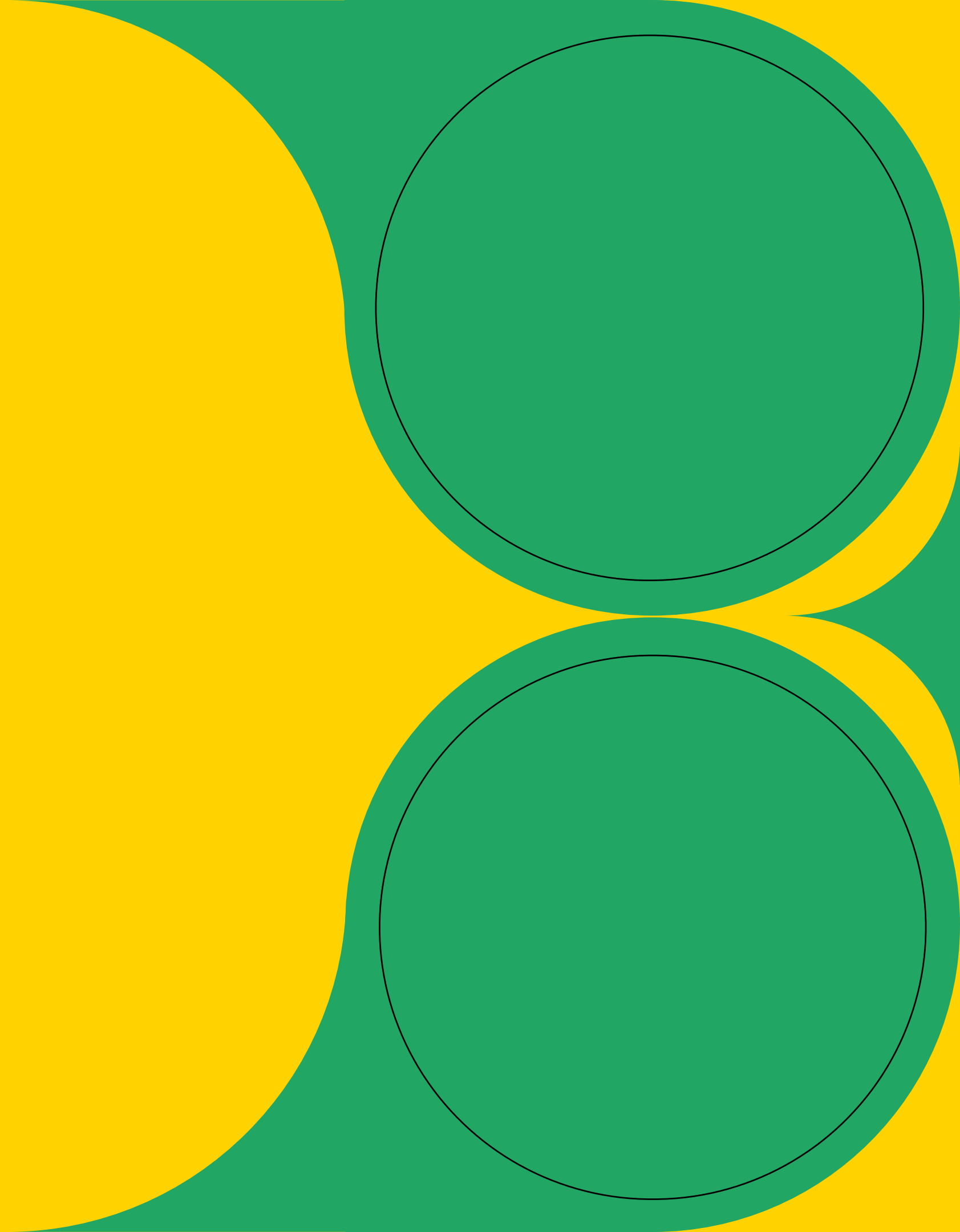


Intelligent Customer Segmentation & Tracking

Building a dynamic system to segment and track customer behavior using RFM analysis.

Goal and Why It Matters

- Build a dynamic system to segment and track customer behavior using RFM analysis.
- Identify high-value customers and improve retention through tailored marketing strategies.
- Track and predict churn customers early.
- Measure campaigns and marketing activities success.



Understanding the RFM Model

- **RFM Analysis** is a method to segment customers based on three key metrics:
 - **Recency (R):** How recently a customer made a purchase.
 - **Frequency (F):** How often they purchase.
 - **Monetary Value (M):** How much they spend.
- Each one alone gives useful insights. For example, if Recency is more than 60 days, we call that customer a *lapse* or *dormant*. If Frequency is high, we might say they're *loyal*.
- But if we only look at one dimension, we miss a lot. To really understand customers, we must combine all three. Together, they give us **70–80% of the customer behavior we need to know**.
- This is an **internal tool for the marketing team** to optimize campaigns and measure their success.

Understanding the RFM Model

Recency is divided into 4 manually assigned bins to fit business context:

- 0–15 days → Score 4 (Most Recent)
- 16–32 days → Score 3(Medium)
- 33–60 days → Score 2(At Risk)
- 60+ days → Score 1 (Dormant)

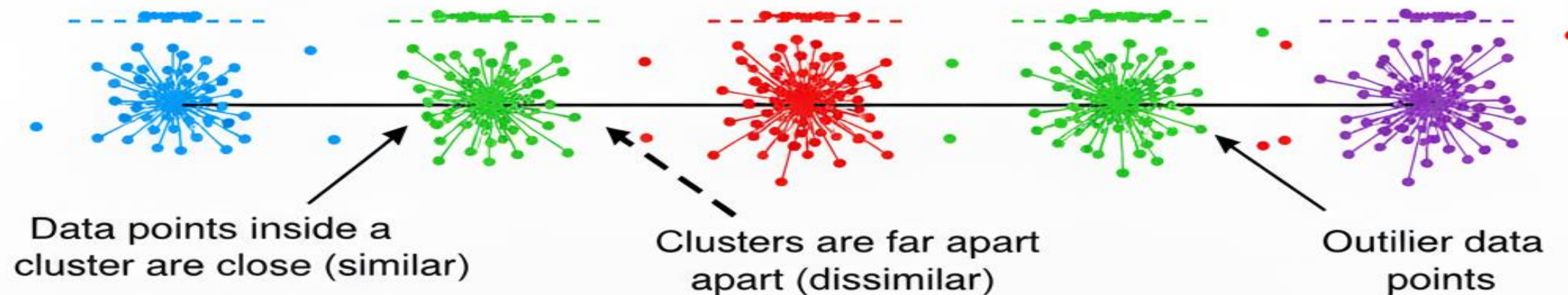
Frequency & Monetary:

- Scored using K-Means Clustering:
 - Groups similar customer behaviors into 4 clusters. 1 (lowest) to 4 (highest).

◆ K-Means = A smart way to let data group itself. It finds natural behavior groups in spending and visits, even if not obvious to humans.

K-means Grouping Visual

K=4 Clusters (1D)

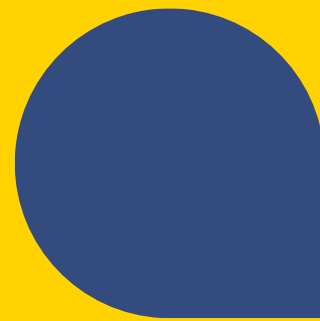


Customer Segmentation Process Explained



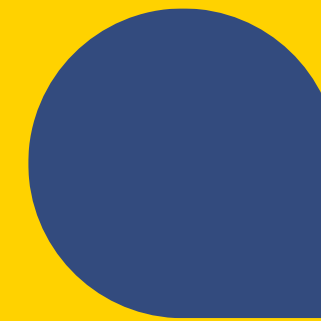
Step 1: Define Segments

Identify key customer segments using RFM analysis.



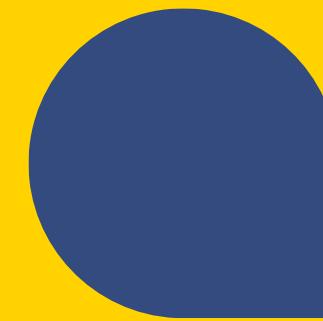
Step 2: Analyze Behavior

Assess customer behavior to tailor marketing strategies effectively.



Step 3: Implement Strategies

Deploy targeted strategies for each customer segment to maximize retention.

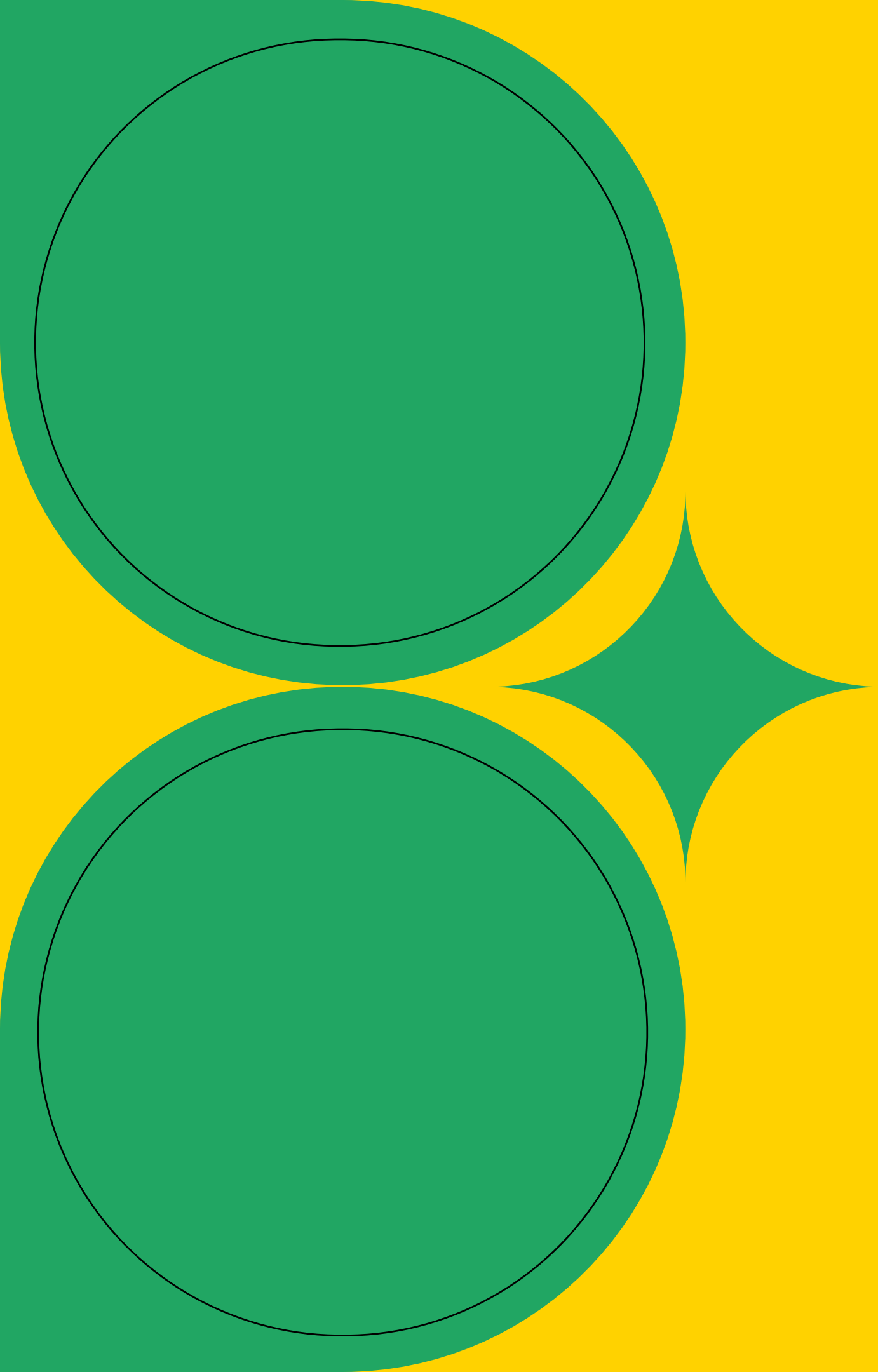


Step 4: Monitor Changes

Continuously track segment movements to refine approaches and improve results.

Strategy	Segments	Explanation	Focus Strategy (Examples)
Grow & Retain	Champion	Top customers: highly engaged, frequent, high spend.	VIP rewards, exclusive deals, cashbacks.
	Loyal	Consistently active & valuable customers.	Loyalty program, upsell, special promos.
	Loyal Low Spender	Frequent visits but low spend per trip.	Basket-size growth: bundles, cross-sell.
	Loyal not frequent	Spend well when active, but not frequent.	Personalized offers to increase visit rate.
Acquire	New	Customers within 30 days of first purchase.	Onboarding: welcome series, first-purchase reward.
Recover/Churn	One-Time Visitor	Made a single purchase, no repeat yet.(Like KITA app)	Win-back: reminder email/WP, discount to return.
	Low Activity	Low frequency, low spend, but recent.	Frequency boost: limited-time deals, reminders.
	At Risk	Declining activity, signs of churn.	Urgent win-back: time-limited offer to return.
	Dormant	No recent activity, possibly lost.	Last-chance: reactivation campaign, big offer.

SEGMENT	R / F / M	EXPLANATION
New	- (≤30 days)	New customer starting their journey.
One Time Visitor	-	Made a single purchase, no repeat yet.
Champion	4/4/4	Highly engaged, frequent, and high spenders. Your top customers.
Loyal	3–4 / 3–4 / 3–4	Consistently active and valuable customers.
Loyal Low Spender	3–4 / 2–4 / 1–2	Shops often but spends little.
Loyal not frequent	3–4 / 1–2 / 3–4	Spends well when active but not frequent.
Low Activity	3–4 / 1–2 / 1–2	Low frequency and low spend, but still recent.
At Risk	2 / X / X	Showing signs of churn. Activity declining.
Dormant	1 / X / X	No recent activity, possibly lost.

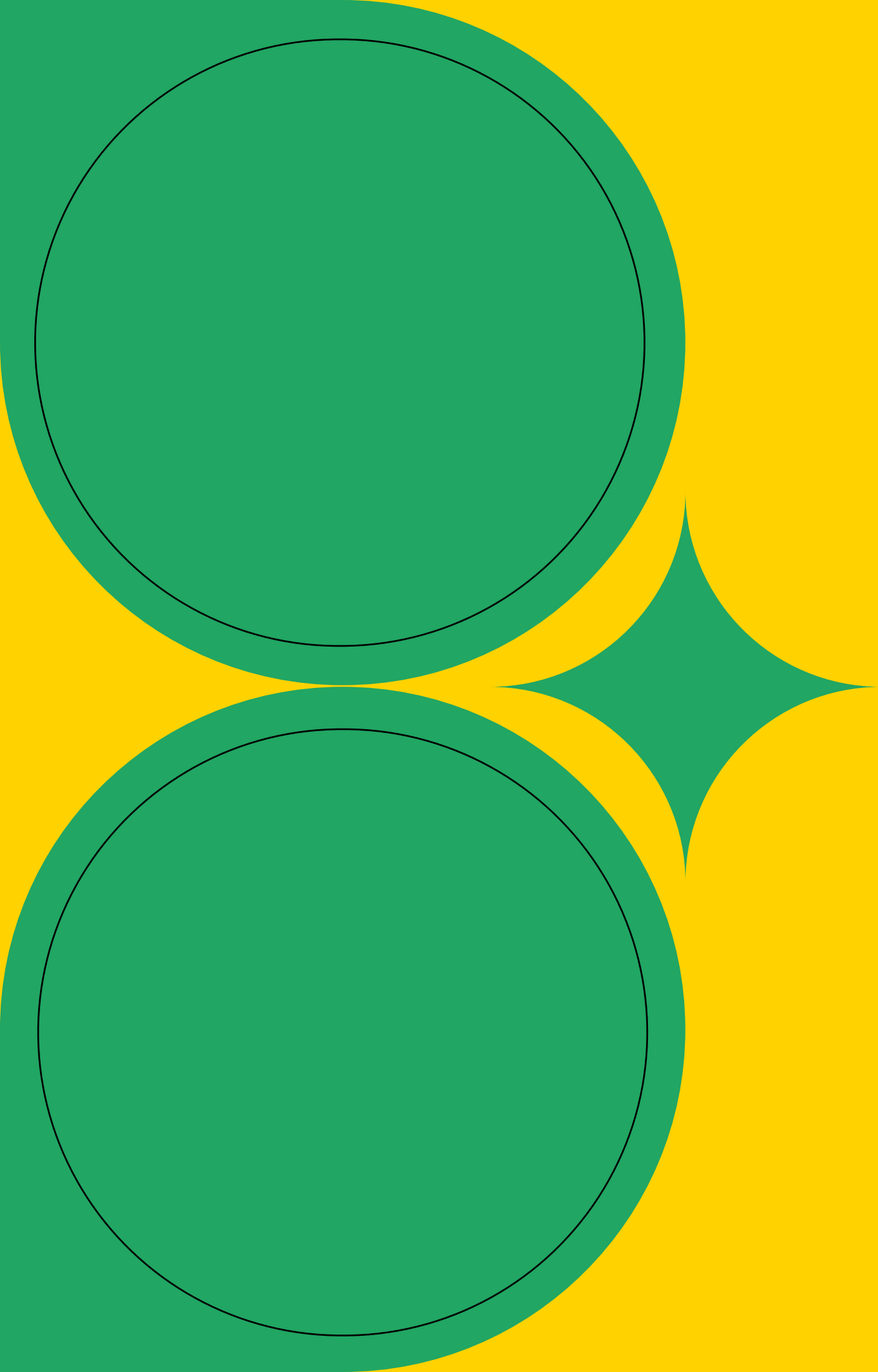


Customer Segments Evolve Over Time

As customer behavior shifts monthly, our model dynamically captures these transitions, helping us implement the right strategies to maintain engagement and optimize marketing efforts.

Example:

- ID111 in July → At Risk
- August → Loyal Low Spender
- September → Loyal



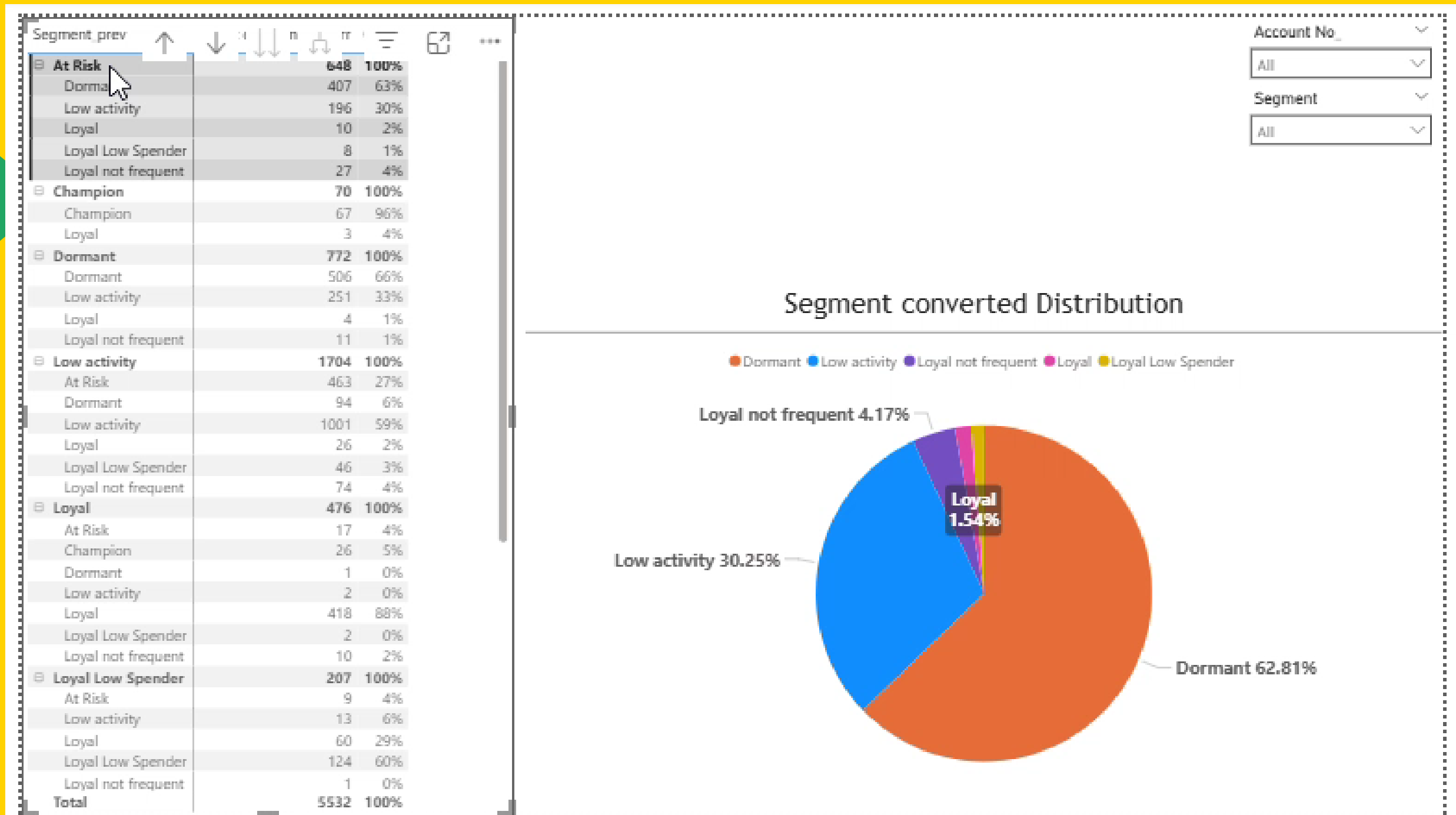
Tracking Customer Segment Movement

This section highlights how customers transition between segments over time. Understanding these shifts allows businesses to tailor their marketing strategies effectively and measure the impact of their campaigns.

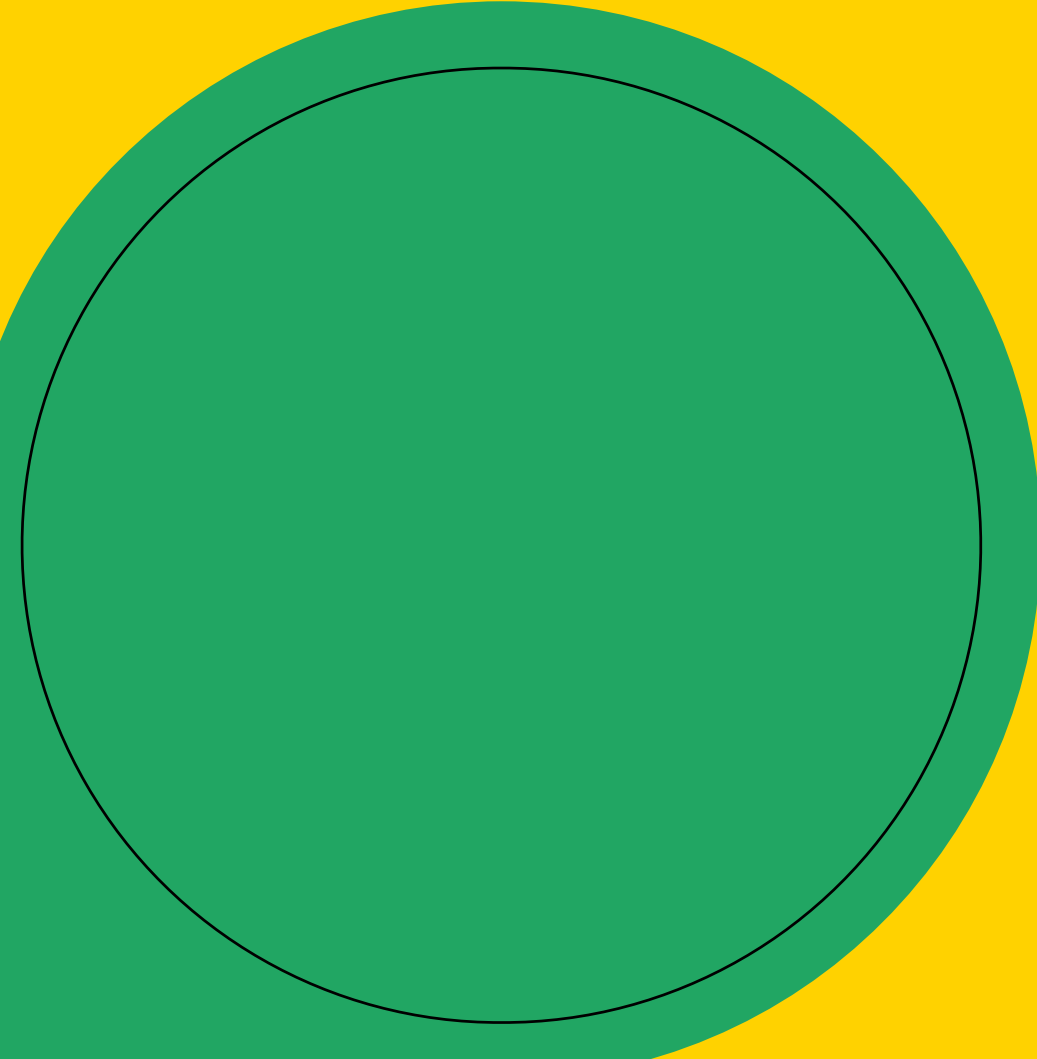
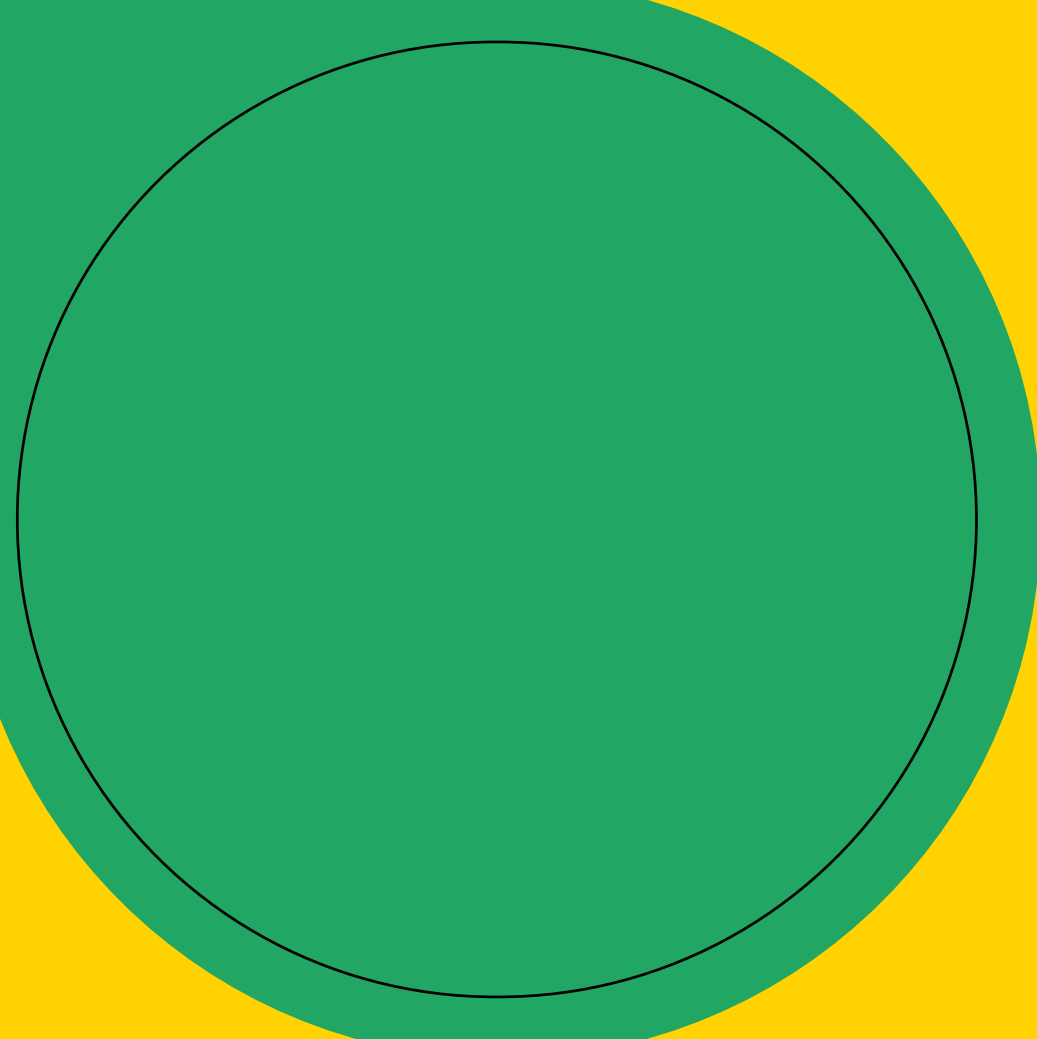
Answers:

- How many At Risk customers became Loyal?
- How many of the new customers last month turned loyal or churned?

Tracking Customer Segment Movement



Actionable Insights for Customer Segmentation



✓ Targeted Campaigns

Launch marketing efforts per segment behavior.

✓ A/B Test Within Segments

Split one segment (e.g., 300 At Risk) into two groups:

- Group A → 5% discount via SMS
- Group B → Voucher on segments they bought from
- Measure which works better.

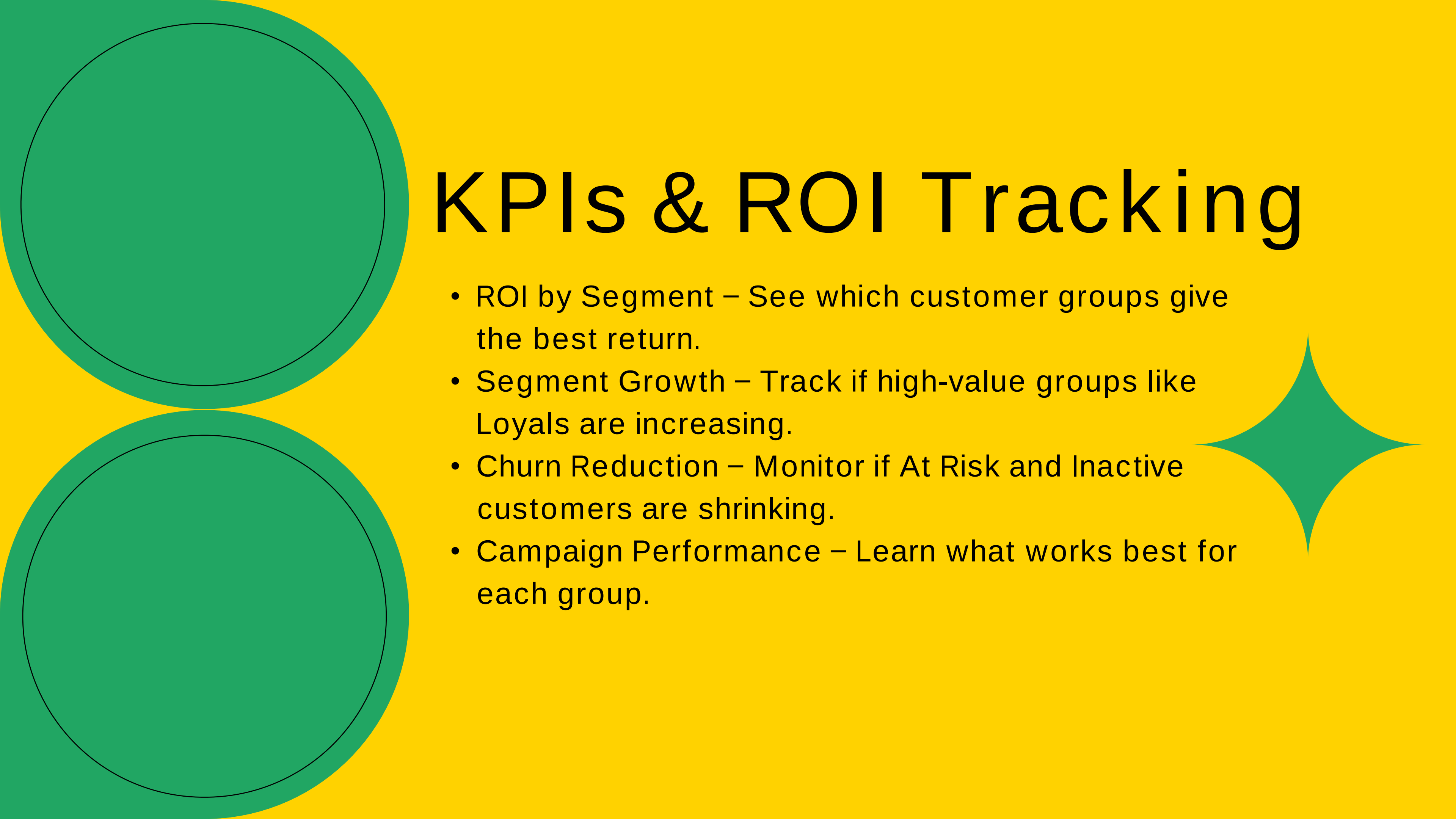
✓ Budget Optimization

Know where ROI comes from — shift spend toward segments that respond.

✓ Campaign Tracking

Monitor month-over-month transitions to measure impact clearly.



The slide features a bright yellow background. On the left side, there are two large, overlapping teal circles. On the right side, there is a teal four-pointed star shape. The title 'KPIs & ROI Tracking' is written in a large, bold, black sans-serif font.

KPIs & ROI Tracking

- ROI by Segment – See which customer groups give the best return.
- Segment Growth – Track if high-value groups like Loyals are increasing.
- Churn Reduction – Monitor if At Risk and Inactive customers are shrinking.
- Campaign Performance – Learn what works best for each group.

Architecture & Flow Overview

- The data flow integrates SQL, Python, and Power BI, ensuring a smooth connection from data cleansing to visualization.
- Each tool serves a crucial role: SQL for live data, Python for scoring, and Power BI for insightful analysis.

